

–Supplemental Material–

Prune but Preserve: Decoupling Redundancy Suppression and Semantic Preservation for Efficient Time Series Clustering

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I. BASELINE DESCRIPTIONS

The baseline set is organized around the empirical question studied in the main paper, namely at which stage redundancy should be reduced for unsupervised MTS clustering. It covers five streams: full-sequence representation learning, input-side token exclusion, partial-observation clustering, structural sub-sequence indexing, and foundation-style time series modeling. This design separates the benefit of source-level patch admission from gains brought by contrastive objectives, spectral modeling, graph structure, incomplete-view learning, or general temporal pretraining. For embedding-output baselines, k -means gives final assignments with the label-provided cluster number k , following the same unsupervised protocol as the main paper.

FCACC [1] is a fuzzy cluster-aware contrastive clustering method for time series. It constructs augmented views and introduces cluster-aware hard negatives to reduce the gap between instance contrast and cluster assignment. FCACC represents a clustering-oriented full-sequence baseline, testing whether cluster-aware contrast can reduce background influence without source-level patch admission. The official implementation is available at [FCACC](#).

TFMCC [2] performs time-frequency augmented multi-level contrastive clustering. Temporal and spectral augmentations enhance cluster-relevant patterns at different representation levels, but the complete sequence still enters the encoding pipeline. TFMCC tests whether temporal-spectral contrast can handle redundancy as effectively as pruning before encoder-side interaction. The official implementation is available at [TFMCC](#).

GTM [3] learns general time series representations through frequency-domain attention and shared temporal modeling. As GTM is designed for transferable representation rather than clustering-specific reduction, k -means is applied to its embeddings for final assignment. The comparison examines whether broad temporal pretraining can replace explicit patch

admission under redundancy-dominated MTS records. The official implementation is available at [GTM](#).

EMTC [4] is the closest input-side exclusion baseline for MTS clustering. It suppresses redundancy through importance-aware variate-wise masking and enriches retained content with multi-endogenous views. Its comparison with SIGMA directly tests whether confidence-calibrated admission, routed semantic restoration, and prototype-guided refinement improve clustering after sparse token admission. The official implementation is available at [EMTC](#).

FEI [5] introduces a frequency-domain information bottleneck by masking frequency components and predicting the corresponding embeddings. This design encourages informative spectral structure without manually defined positive and negative pairs. FEI serves as a spectral-reduction baseline, assessing whether patch-level admission still helps beyond frequency-oriented representation learning. The official implementation is available at [FEI](#).

MVCIMTS [6] is a contrastive multi-view clustering method for MTS. It treats variates as views and learns complementary information through multi-view contrast. Since its partial observation comes from variables instead of temporal patches, MVCIMTS tests whether variable-wise incompleteness brings the same clustering benefit as temporal admission followed by routed restoration. The official implementation is available at [MVCIMTS](#).

k -Graph [7] converts subsequences into graph representations and clusters graph-derived features. By building graphs across multiple subsequence lengths, k -Graph captures recurring local patterns and structural similarity. It provides a data-centric baseline for datasets whose clusters depend on shape correspondence or repeated subsequence structure. The official implementation is available at [k-Graph](#).

TimesURL [8] is a universal self-supervised representation method for time series. It combines contrastive learning with temporal constraints to obtain general embeddings. In the prune-and-preserve comparison, TimesURL represents a

full-sequence self-supervised baseline because redundancy is not controlled through an admissible patch set. The official implementation is available at [TimesURL](#).

UNITS [9] is a unified multi-task time series model. It represents task information as tokens in a shared modeling framework and learns transferable representations across domains, temporal scales, and sampling patterns. UNITS provides a foundation-style baseline for testing whether task-unified pre-training is sufficient for redundancy-dominated MTS clustering. Its comparison with SIGMA highlights the role of source-level admission and routed semantic preservation in cluster-oriented representation learning. The official implementation is available at [UNITS](#).

II. IMPLEMENTATION DETAILS

All experiments follow the unsupervised clustering protocol in the main paper. For each UEA dataset, training and test splits are concatenated before clustering, and labels are accessed only for metric computation. Each experiment is repeated with five random seeds, and the mean and standard deviation are reported. This protocol keeps the supplemental material consistent with the main-paper evaluation and makes the supplementary tables comparable to the representative results in Section IV of the full paper.

SIGMA uses Confidence-Calibrated Masked Admission as the default admission strategy. In the masking ablation, source-level admission is removed while all other settings remain unchanged, isolating the effect of controlling patch participation before encoder-side feature interaction. Unless otherwise specified, no input perturbation is added, and Gaussian noise is introduced only in the robustness study.

For patch-level admission, the patch length is set to $\ell_p = 4$. Given $m = \lceil T/\ell_p \rceil$ patches, the retained ratio is $r_a = 0.5$, and the retained patch number is $m_a = \lceil r_a m \rceil$. Thus, the encoder processes a compact patch subset after admission. This setting reduces redundant patch participation before feature interaction while keeping enough temporal context for semantic restoration.

For endogenous-view representation learning, the number of views is $n_v = 2$. Each view encoder uses hidden dimension $d_h = 32$, and the fused clustering embedding has dimension $d_z = 64$. The prototype alignment temperature is $\tau_p = 0.1$. These values give SIGMA a compact representation budget while allowing admitted content to be organized into complementary latent views.

SIGMA is trained for $e_{\max} = 200$ epochs with mini-batch size $b = 64$. Adam is adopted with learning rate $\alpha = 0.001$. The loss weights are $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{orth}} = 0.1$, and $\lambda_{\text{pa}} = 1.0$, corresponding to reconstruction, inter-view orthogonality, and prototype alignment, respectively. Reconstruction anchors temporal content after admission, orthogonality discourages view homogenization, and prototype alignment sharpens the fused cluster space. Under this configuration, admission control, semantic restoration, and cluster refinement keep distinct roles within the prune-and-preserve pipeline.

III. SUPPLEMENTARY RESULTS

This section provides the full experimental record behind the condensed evaluation in the main paper. All results follow the unsupervised clustering protocol in Section II, so the added tables and figures serve as expanded checks rather than separate evaluation settings. The supplementary material includes complete 30-dataset results, Bonferroni-Dunn rank tests, detailed architecture and route ablations, loss-evolution curves, admission-based scalability tests, and additional robustness and compactness diagnostics.

The presentation follows the experimental questions in the main paper. Complete benchmark tables and rank tests extend Q1 from representative datasets to the full UEA benchmark. Architecture and route-internal ablations expand Q2 by tracing the roles of admission, view generation, reconstruction routing, and prototype refinement. Larger-scale and mask-rate tests further examine Q3, and the additional robustness diagnostics complement Q4 under noise perturbation and embedding compression. This structure lets readers connect each supplementary result to a specific claim in the main text without changing the evaluation protocol.

A. Complete Clustering Results on 30 UEA Datasets

TABLE I and TABLE II report the complete Q1 results on all 30 UEA datasets. ACC, F1, NMI, and ARI are computed under the same unsupervised protocol as the main paper. This subsection checks whether the trends reported on representative datasets extend to the full benchmark.

The complete records show that SIGMA performs well on datasets where cluster-related changes appear in compact fragments. For example, SIGMA reaches 0.9875 ACC and 0.9666 ARI on BasicMotions, obtains 0.8525 ACC and 0.7578 ARI on JapaneseVowels, and improves ACC and ARI to 0.6841 and 0.5305 on PenDigits. On SelfRegulationSCP1, SIGMA also gives the best ACC, F1, and ARI. These cases are consistent with the main-paper claim that reducing patch participation before encoding can preserve useful grouping cues.

The full tables also show that different datasets still favor different modeling biases. FCACC performs well on CharacterTrajectories, Cricket, NATOPS, and UWaveGestureLibrary. EMTC remains competitive on AtrialFibrillation, FingerMovements, HandMovementDirection, and MotorImagery. TFMCC leads several shape-sensitive or spectral-sensitive cases, including DuckDuckGeese, ERing, and RacketSports, and FEI performs well on Epilepsy, Libras, and PhonemeSpectra. Therefore, the full benchmark does not suggest that one design dominates every temporal pattern. Instead, Fig. 1 summarizes the aggregate behavior, where SIGMA obtains the lowest average rank on ACC, F1, NMI, and ARI. The complete tables and the rank test together show that the main-paper results are not limited to selected examples.

B. Complete Architecture and Route Ablations

The full paper reports averaged ablation results for Q2. The supplementary tables provide dataset-level results on all 30

TABLE I
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART I. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND. OOM DENOTES OUT-OF-MEMORY RUNS.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
ArticularyWordRecognition	ACC	0.7329 ± 0.0037	0.3555 ± 0.0380	0.3547 ± 0.0185	0.5030 ± 0.0253	0.4710 ± 0.0731	0.0497 ± 0.0124	0.2838 ± 0.0159	0.6929 ± 0.0516	0.2948 ± 0.0200	0.7603 ± 0.0232
	F1	0.8039 ± 0.0055	0.3526 ± 0.0348	0.3560 ± 0.0214	0.3994 ± 0.0241	0.4974 ± 0.0815	0.0109 ± 0.0096	0.2774 ± 0.0169	0.7425 ± 0.0413	0.3114 ± 0.0243	0.7929 ± 0.0392
	NMI	0.8408 ± 0.0064	0.4708 ± 0.0311	0.5070 ± 0.0272	0.6185 ± 0.0159	0.6185 ± 0.0744	0.0314 ± 0.0428	0.3948 ± 0.0099	0.8215 ± 0.0145	0.4280 ± 0.0198	0.8276 ± 0.0165
	ARI	0.6641 ± 0.0045	0.1833 ± 0.0296	0.1615 ± 0.0271	0.4466 ± 0.0210	0.3352 ± 0.0843	0.0019 ± 0.0031	0.1224 ± 0.0091	0.6214 ± 0.0340	0.1395 ± 0.0188	0.6466 ± 0.0257
AtrialFibrillation	ACC	0.4333 ± 0.0000	0.4667 ± 0.0422	0.6133 ± 0.0499	0.4333 ± 0.0558	0.3867 ± 0.0163	0.4095 ± 0.0440	0.4400 ± 0.0389	0.4600 ± 0.0490	0.4500 ± 0.0167	0.5000 ± 0.0365
	F1	0.4303 ± 0.0028	0.4593 ± 0.0393	0.5828 ± 0.0633	0.4041 ± 0.0191	0.3543 ± 0.0194	0.3491 ± 0.0555	0.4295 ± 0.0406	0.4169 ± 0.0278	0.4343 ± 0.0089	0.4956 ± 0.0778
	NMI	0.1005 ± 0.0040	0.0726 ± 0.0277	0.2883 ± 0.0758	0.0544 ± 0.0351	0.0353 ± 0.0237	0.1445 ± 0.0821	0.1143 ± 0.0748	0.0858 ± 0.0339	0.0643 ± 0.0072	0.1251 ± 0.0390
	ARI	-0.0056 ± 0.0011	0.0051 ± 0.0258	0.1248 ± 0.0788	0.0064 ± 0.0282	-0.0360 ± 0.0089	0.1058 ± 0.0600	0.0039 ± 0.0462	0.0136 ± 0.0342	-0.0125 ± 0.0012	0.0582 ± 0.0393
BasicMotions	ACC	0.6675 ± 0.0100	0.7300 ± 0.1086	0.7950 ± 0.0292	0.9850 ± 0.0122	0.6775 ± 0.0443	0.2975 ± 0.0827	0.8875 ± 0.1321	0.7700 ± 0.1582	0.5437 ± 0.0812	0.9875 ± 0.0079
	F1	0.8228 ± 0.0098	0.7311 ± 0.1060	0.7832 ± 0.0275	0.9701 ± 0.0235	0.7351 ± 0.0723	0.1517 ± 0.0794	0.8792 ± 0.1500	0.8627 ± 0.0835	0.5705 ± 0.0412	0.9875 ± 0.0079
	NMI	0.8048 ± 0.0020	0.5569 ± 0.1633	0.6949 ± 0.0711	0.9627 ± 0.0248	0.6792 ± 0.0317	0.1007 ± 0.1775	0.8626 ± 0.0575	0.8487 ± 0.0634	0.3898 ± 0.1107	0.9665 ± 0.0191
	ARI	0.6295 ± 0.0045	0.4526 ± 0.1902	0.6085 ± 0.0606	0.9606 ± 0.0311	0.5317 ± 0.0351	0.0690 ± 0.1381	0.8174 ± 0.1218	0.7349 ± 0.1448	0.2989 ± 0.1276	0.9666 ± 0.0208
CharacterTrajectories	ACC	0.7981 ± 0.0012	0.0651 ± 0.0000	0.3000 ± 0.0261	0.3677 ± 0.0448	0.6040 ± 0.0096	0.0866 ± 0.0318	0.4500 ± 0.0167	0.0651 ± 0.0000	0.4533 ± 0.0267	0.7668 ± 0.0529
	F1	0.8583 ± 0.0012	0.0061 ± 0.0000	0.2886 ± 0.0434	0.3405 ± 0.0281	0.6289 ± 0.0099	0.0196 ± 0.0166	0.4350 ± 0.0149	0.1222 ± 0.0000	0.1703 ± 0.0373	0.8019 ± 0.0378
	NMI	0.8726 ± 0.0013	0.0000 ± 0.0000	0.3633 ± 0.0432	0.5756 ± 0.0418	0.7095 ± 0.0041	0.0682 ± 0.1107	0.5347 ± 0.0121	0.0000 ± 0.0000	0.1776 ± 0.0688	0.8261 ± 0.0315
	ARI	0.7705 ± 0.0012	0.0000 ± 0.0000	0.1756 ± 0.0296	0.1913 ± 0.0474	0.5137 ± 0.0138	0.0181 ± 0.0345	0.3211 ± 0.0193	0.0000 ± 0.0000	0.0532 ± 0.0279	0.6933 ± 0.0563
Cricket	ACC	0.9407 ± 0.0288	0.4533 ± 0.0264	0.5528 ± 0.0204	0.6922 ± 0.0664	0.5744 ± 0.0834	0.1167 ± 0.0350	0.4589 ± 0.0218	0.6722 ± 0.0813	0.4922 ± 0.0990	0.8489 ± 0.0787
	F1	0.9599 ± 0.0024	0.4393 ± 0.0376	0.5635 ± 0.0383	0.6319 ± 0.0703	0.6027 ± 0.0590	0.0417 ± 0.0250	0.4545 ± 0.0225	0.7562 ± 0.0557	0.5414 ± 0.0992	0.8838 ± 0.0523
	NMI	0.9509 ± 0.0020	0.5189 ± 0.0101	0.6181 ± 0.0286	0.7858 ± 0.0464	0.6794 ± 0.0579	0.0817 ± 0.0964	0.4752 ± 0.0228	0.7958 ± 0.0391	0.5947 ± 0.0841	0.8764 ± 0.0455
	ARI	0.9051 ± 0.0158	0.2584 ± 0.0194	0.2971 ± 0.0365	0.5842 ± 0.0827	0.4468 ± 0.0872	0.0171 ± 0.0299	0.2533 ± 0.0250	0.6028 ± 0.0783	0.3359 ± 0.1072	0.7672 ± 0.0967
DuckDuckGeese	ACC	0.3067 ± 0.0125	0.4140 ± 0.0162	0.4240 ± 0.0408	0.4720 ± 0.0172	0.3220 ± 0.0117	0.2620 ± 0.0519	0.3280 ± 0.0160	0.3220 ± 0.0147	0.2960 ± 0.0136	0.4020 ± 0.0325
	F1	0.2611 ± 0.0200	0.4042 ± 0.0246	0.4003 ± 0.0494	0.3394 ± 0.0193	0.3013 ± 0.0347	0.1650 ± 0.0575	0.3017 ± 0.0107	0.2641 ± 0.0182	0.2869 ± 0.0075	0.3819 ± 0.0291
	NMI	0.0800 ± 0.0179	0.2483 ± 0.0205	0.2082 ± 0.0380	0.2791 ± 0.0267	0.1030 ± 0.0090	0.0335 ± 0.0301	0.0958 ± 0.0101	0.1732 ± 0.0351	0.0682 ± 0.0175	0.1718 ± 0.0577
	ARI	0.0227 ± 0.0109	0.1394 ± 0.0210	0.0729 ± 0.0331	0.1780 ± 0.0248	0.0369 ± 0.0091	0.0073 ± 0.0170	0.0225 ± 0.0087	0.0375 ± 0.0110	0.0064 ± 0.0105	0.1000 ± 0.0580
EigenWorms	ACC	OOM	0.3876 ± 0.0072	OOM	OOM	0.4185 ± 0.0361	0.4324 ± 0.0024	0.3135 ± 0.0236	0.3660 ± 0.0510	0.4525 ± 0.0366	0.4332 ± 0.0402
	F1	OOM	0.2868 ± 0.0068	OOM	OOM	0.3899 ± 0.0605	0.1545 ± 0.0173	0.2920 ± 0.0180	0.3035 ± 0.0806	0.4410 ± 0.0601	0.4016 ± 0.0483
	NMI	OOM	0.0683 ± 0.0035	OOM	OOM	0.1826 ± 0.0152	0.0300 ± 0.0094	0.0848 ± 0.0181	0.0697 ± 0.0572	0.2872 ± 0.0919	0.1885 ± 0.0300
	ARI	OOM	0.0626 ± 0.0052	OOM	OOM	0.1453 ± 0.0227	0.0098 ± 0.0046	0.0400 ± 0.0160	0.0515 ± 0.0699	0.1694 ± 0.0486	0.1620 ± 0.0458
Epilepsy	ACC	0.4279 ± 0.0297	0.3869 ± 0.0404	0.5203 ± 0.0244	0.4953 ± 0.0477	0.6364 ± 0.0808	0.2953 ± 0.0324	0.6167 ± 0.0316	0.5156 ± 0.0189	0.4982 ± 0.0646	0.4756 ± 0.0321
	F1	0.4295 ± 0.0302	0.3710 ± 0.0430	0.5136 ± 0.0265	0.4147 ± 0.0552	0.6200 ± 0.0821	0.1882 ± 0.0235	0.6200 ± 0.0260	0.4979 ± 0.0080	0.5086 ± 0.0930	0.4724 ± 0.0357
	NMI	0.1397 ± 0.0347	0.1015 ± 0.0345	0.2476 ± 0.0419	0.3029 ± 0.0794	0.4079 ± 0.0773	0.0205 ± 0.0340	0.4013 ± 0.0415	0.3363 ± 0.0795	0.3126 ± 0.0851	0.2652 ± 0.0289
	ARI	0.0960 ± 0.0211	0.0814 ± 0.0434	0.1803 ± 0.0389	0.1992 ± 0.0638	0.3829 ± 0.0804	0.0076 ± 0.0184	0.3275 ± 0.0526	0.2288 ± 0.0542	0.2320 ± 0.0733	0.1889 ± 0.0197
ERing	ACC	0.6327 ± 0.0360	0.4180 ± 0.0585	0.6704 ± 0.0486	0.8607 ± 0.0218	0.7120 ± 0.0918	0.3667 ± 0.1560	0.5380 ± 0.0421	0.7433 ± 0.1161	0.4087 ± 0.0639	0.7980 ± 0.0749
	F1	0.6306 ± 0.0393	0.4117 ± 0.0586	0.6741 ± 0.0496	0.7564 ± 0.0332	0.7508 ± 0.0811	0.2682 ± 0.1646	0.5469 ± 0.0403	0.7862 ± 0.0839	0.4159 ± 0.0698	0.8003 ± 0.0701
	NMI	0.5660 ± 0.0104	0.2357 ± 0.0481	0.5487 ± 0.0473	0.7821 ± 0.0321	0.6562 ± 0.0772	0.3054 ± 0.2115	0.3843 ± 0.0143	0.7407 ± 0.0313	0.2528 ± 0.0802	0.6782 ± 0.0892
	ARI	0.4231 ± 0.0204	0.1602 ± 0.0466	0.4382 ± 0.0499	0.7051 ± 0.0415	0.5669 ± 0.0959	0.1897 ± 0.1599	0.2816 ± 0.0319	0.6467 ± 0.0757	0.1578 ± 0.0647	0.6168 ± 0.1270
EthanolConcentration	ACC	0.2992 ± 0.0025	0.2786 ± 0.0051	0.3605 ± 0.0224	0.2634 ± 0.0096	0.2901 ± 0.0017	0.2962 ± 0.0172	0.2920 ± 0.0100	0.2962 ± 0.0044	0.2691 ± 0.0034	0.2969 ± 0.0153
	F1	0.2759 ± 0.0027	0.2529 ± 0.0093	0.3458 ± 0.0279	0.4681 ± 0.0189	0.2744 ± 0.0087	0.2442 ± 0.0306	0.2882 ± 0.0110	0.2747 ± 0.0067	0.2090 ± 0.0059	0.2732 ± 0.0102
	NMI	0.0149 ± 0.0009	0.0047 ± 0.0010	0.0574 ± 0.0189	0.0124 ± 0.0098	0.0116 ± 0.0018	0.0176 ± 0.0112	0.0074 ± 0.0029	0.0199 ± 0.0025	0.0025 ± 0.0015	0.0190 ± 0.0157
	ARI	0.0012 ± 0.0003	-0.0013 ± 0.0010	0.0307 ± 0.0121	0.0003 ± 0.0007	0.0032 ± 0.0005	0.0072 ± 0.0073	0.0008 ± 0.0024	0.0039 ± 0.0009	-0.0014 ± 0.0010	0.0097 ± 0.0119
FaceDetection	ACC	0.5130 ± 0.0030	OOM	0.5146 ± 0.0047	0.5051 ± 0.0029	0.5077 ± 0.0031	OOM	0.5038 ± 0.0018	0.5001 ± 0.0000	0.5054 ± 0.0040	0.5198 ± 0.0044
	F1	0.5125 ± 0.0028	OOM	0.4808 ± 0.0369	0.6113 ± 0.0585	0.5051 ± 0.0039	OOM	0.5028 ± 0.0023	0.3336 ± 0.0000	0.5030 ± 0.0060	0.5192 ± 0.0046
	NMI	0.0005 ± 0.0002	OOM	0.0010 ± 0.0004	0.0002 ± 0.0002	0.0002 ± 0.0001	OOM	0.0001 ± 0.0000	0.0027 ± 0.0000	0.0001 ± 0.0002	0.0012 ± 0.0005
	ARI	0.0006 ± 0.0003	OOM	0.0007 ± 0.0006	0.0001 ± 0.0001	0.0002 ± 0.0002	OOM	-0.0000 ± 0.0000	0.0000 ± 0.0000	0.0001 ± 0.0003	0.0015 ± 0.0006
FingerMovements	ACC	0.5250 ± 0.0052	0.5183 ± 0.0085	0.5680 ± 0.0160	0.5072 ± 0.0055	0.5163 ± 0.0127	0.5260 ± 0.0041	0.5264 ± 0.0026	0.5192 ± 0.0082	0.5130 ± 0.0054	0.5221 ± 0.0129
	F1	0.4951 ± 0.0111	0.4982 ± 0.0212	0.5349 ± 0.0218	0.6004 ± 0.0599	0.5051 ± 0.0184	0.5230 ± 0.0023	0.4997 ± 0.0023	0.5161 ± 0.0058	0.5124 ± 0.0051	0.5183 ± 0.0074
	NMI	0.0027 ± 0.0009	0.0014 ± 0.0009	0.0210 ± 0.0132	0.0023 ± 0.0035	0.0013 ± 0.0016	0.0021 ± 0.0007	0.0028 ± 0.0005	0.0013 ± 0.0009	0.0006 ± 0.0003	0.0022 ± 0.0026
	ARI	0.0008 ± 0.0009	-0.0004 ± 0.0009	0.0121 ± 0.0087	-0.0010 ± 0.0008	-0.0005 ± 0.0021	0.0004 ± 0.0009	0.0009 ± 0.0005	-0.0006 ± 0.0013	-0.0016 ± 0.0004	0.0003 ± 0.0031
HandMovementDirection	ACC	0.2829 ± 0.0063	0.2957 ± 0.0087	0.4405 ± 0.0359	0.3248 ± 0.0164	0.3017 ± 0.0088	0.3017 ± 0.0123	0.3026 ± 0.0099	0.3120 ± 0.0097	0.3743 ± 0.0375	0.3179 ± 0.0100
	F1	0.2108 ± 0.0397	0.2907 ± 0.0061	0.4070 ± 0.0226	0.3328 ± 0.0189	0.2867 ± 0.0112	0.2667 ± 0.0205	0.2996 ± 0.0128	0.3082 ± 0.0138	0.3667 ± 0.0383	0.3145 ± 0.0104
	NMI	0.0168 ± 0.0104	0.0156 ± 0.0061	0.0910 ± 0.0245	0.0286 ± 0.0110	0.0184 ± 0.0049	0.0206 ± 0.0103	0.0144 ± 0.0047	0.0208 ± 0.0045	0.0818 ± 0.0394	0.0208 ± 0.0050
	ARI	-0.0047 ± 0.0038	0.0015 ± 0.0062	0.0475 ± 0.0166	0.0109 ± 0.0104	0.0001 ± 0.0045	0.0007 ± 0.0032	0.0003 ± 0.0032	0.0046 ± 0.0035	0.0488 ± 0.0282	0.0052 ± 0.0047
Handwriting	ACC	0.2433 ± 0.0041	0.1264 ± 0.0027	0.1536 ± 0.0097	0.1090 ± 0.0209	0.1778 ± 0.0097	0.0878 ± 0.0160	0.1708 ± 0.0101	0.1488 ± 0.0067	0.2940 ± 0.0068	0.2530 ± 0.0126
	F1	0.2418 ± 0.0080	0.1260 ± 0.0020	0.1501 ± 0.0114	0.1206 ± 0.0171	0.1741 ± 0.0071	0.0352 ± 0.0127	0.1679 ± 0.0098	0.1484 ± 0.0078	0.2881 ± 0.0075	0.2591 ± 0.0132
	NMI	0.3723 ± 0.0061	0.1523 ± 0.0052	0.1934 ± 0.0070	0.1228 ± 0.0510	0.2497 ± 0.0235	0.0637 ± 0.0274	0.2260 ± 0.0097	0.2009 ± 0.0035	0.0101 ± 0.0049	0.3652 ± 0.0139
	ARI	0.1106 ± 0.0035	0.0125 ± 0.0016	0.0254 ± 0.0036	0.0195 ± 0.0090	0.0494 ± 0.0107	0.0084 ± 0.0048	0.0429 ± 0.0045	0.0282 ± 0.0012	-0.0047 ± 0.0030	0.1134 ± 0.0113

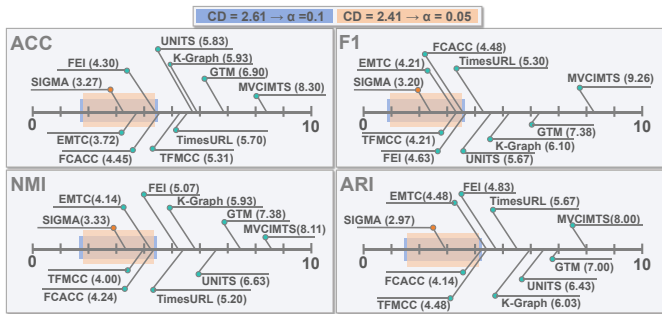


Fig. 1. Bonferroni-Dunn rank test on 30 UEA datasets under ACC, F1, NMI, and ARI. Lower average rank indicates stronger clustering quality, and critical-difference bars report significance thresholds at $\alpha = 0.1$ and $\alpha = 0.05$.

UEA datasets under the protocol in Section II. The architecture ablations remove CCMA and MEVG separately or together, examining whether early patch admission and endogenous-view representation each contribute to SIGMA. The route ablations remove the reconstruction route and the contrastive route separately or together, examining whether temporal recovery and prototype-oriented grouping remain useful after admission.

TABLE III and TABLE IV report the architecture results. The complete SIGMA model gives the best overall average, although several datasets favor a reduced variant on individual metrics

TABLE II
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART II. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
Heartbeat	ACC	0.5844 ± 0.0079	0.5340 ± 0.0292	0.7180 ± 0.0020	0.7076 ± 0.0476	0.7090 ± 0.0000	0.6758 ± 0.0497	0.7296 ± 0.0025	0.7149 ± 0.0106	0.6244 ± 0.0238	0.6817 ± 0.0546
	F1	0.5688 ± 0.0055	0.4974 ± 0.0193	0.7553 ± 0.1598	0.6934 ± 0.0669	0.8298 ± 0.0000	0.4592 ± 0.0483	0.5461 ± 0.0071	0.7922 ± 0.0760	0.5347 ± 0.0192	0.5944 ± 0.0367
	NMI	0.0389 ± 0.0016	0.0012 ± 0.0015	0.0106 ± 0.0023	0.0465 ± 0.0073	0.0204 ± 0.0000	0.0016 ± 0.0020	0.0344 ± 0.0041	0.0300 ± 0.0207	0.0070 ± 0.0077	0.0382 ± 0.0220
	ARI	0.0258 ± 0.0062	0.0030 ± 0.0062	-0.0029 ± 0.0061	0.0966 ± 0.0306	-0.1144 ± 0.0000	0.0018 ± 0.0101	0.0875 ± 0.0071	0.0220 ± 0.0714	0.0243 ± 0.0111	0.0981 ± 0.0485
InsectWingbeat	ACC	0.1651 ± 0.0020	0.1003 ± 0.0000	0.1251 ± 0.0049	0.1099 ± 0.0024	0.1533 ± 0.0004	OOM	0.1451 ± 0.0010	0.1000 ± 0.0000	0.0872 ± 0.0071	0.1336 ± 0.0017
	F1	0.1623 ± 0.0020	0.0189 ± 0.0001	0.0945 ± 0.0222	0.0962 ± 0.0074	0.1062 ± 0.0051	OOM	0.1374 ± 0.0049	0.1818 ± 0.0000	0.0812 ± 0.0055	0.1209 ± 0.0043
	NMI	0.0428 ± 0.0011	0.0010 ± 0.0001	0.0090 ± 0.0015	0.1883 ± 0.0060	0.0339 ± 0.0008	OOM	0.0210 ± 0.0008	0.0000 ± 0.0000	0.0032 ± 0.0061	0.0146 ± 0.0015
	ARI	0.0216 ± 0.0005	0.0000 ± 0.0000	0.0027 ± 0.0013	0.0377 ± 0.0055	0.0130 ± 0.0007	OOM	0.0087 ± 0.0002	0.0000 ± 0.0000	0.0043 ± 0.0000	0.0059 ± 0.0006
JapaneseVowels	ACC	0.6286 ± 0.0530	0.1844 ± 0.0000	0.3203 ± 0.0273	0.4766 ± 0.0639	0.5198 ± 0.0223	0.1844 ± 0.0000	0.3369 ± 0.0199	0.1844 ± 0.0000	0.2619 ± 0.0271	0.8525 ± 0.0431
	F1	0.6617 ± 0.0303	0.0346 ± 0.0000	0.3325 ± 0.0365	0.4274 ± 0.0355	0.5551 ± 0.0053	0.0346 ± 0.0000	0.3069 ± 0.0236	0.3113 ± 0.0000	0.2755 ± 0.0282	0.8494 ± 0.0383
	NMI	0.5950 ± 0.0134	0.0000 ± 0.0000	0.2410 ± 0.0418	0.5250 ± 0.0397	0.4727 ± 0.0084	0.0000 ± 0.0000	0.2820 ± 0.0112	0.0000 ± 0.0000	0.1643 ± 0.0209	0.8113 ± 0.0180
	ARI	0.4771 ± 0.0389	0.0000 ± 0.0000	0.1269 ± 0.0412	0.3167 ± 0.0466	0.3049 ± 0.0281	0.0000 ± 0.0000	0.1511 ± 0.0210	0.0000 ± 0.0000	0.0648 ± 0.0171	0.7578 ± 0.0423
Libras	ACC	0.3822 ± 0.0052	0.3894 ± 0.0321	0.4011 ± 0.0315	0.3544 ± 0.0099	0.4317 ± 0.0256	0.1456 ± 0.0416	0.3844 ± 0.0208	0.2700 ± 0.0148	0.2839 ± 0.0116	0.3711 ± 0.0125
	F1	0.3931 ± 0.0193	0.3772 ± 0.0369	0.4028 ± 0.0406	0.2626 ± 0.0226	0.4438 ± 0.0298	0.0627 ± 0.0302	0.3700 ± 0.0242	0.2767 ± 0.0166	0.2853 ± 0.0157	0.3655 ± 0.0158
	NMI	0.5311 ± 0.0095	0.4697 ± 0.0327	0.4890 ± 0.0322	0.4698 ± 0.0245	0.5056 ± 0.0189	0.1311 ± 0.0783	0.5126 ± 0.0062	0.3101 ± 0.0185	0.3562 ± 0.0152	0.4086 ± 0.0259
	ARI	0.2484 ± 0.0082	0.2143 ± 0.0276	0.1881 ± 0.0295	0.2019 ± 0.0228	0.2423 ± 0.0142	0.0370 ± 0.0255	0.2532 ± 0.0127	0.0901 ± 0.0166	0.1246 ± 0.0146	0.2809 ± 0.0196
LSST	ACC	0.3434 ± 0.0022	0.1224 ± 0.0038	0.3672 ± 0.0074	0.2491 ± 0.0169	0.3327 ± 0.0037	0.3227 ± 0.0092	0.1578 ± 0.0113	0.3293 ± 0.0025	0.1805 ± 0.0098	0.2019 ± 0.0029
	F1	0.1918 ± 0.0134	0.0984 ± 0.0072	0.2055 ± 0.0297	0.2114 ± 0.0148	0.1689 ± 0.0141	0.0521 ± 0.0222	0.1149 ± 0.0053	0.2387 ± 0.0157	0.1376 ± 0.0084	0.1840 ± 0.0147
	NMI	0.1188 ± 0.0020	0.0311 ± 0.0056	0.1959 ± 0.0261	0.2066 ± 0.0198	0.1338 ± 0.0040	0.0237 ± 0.0303	0.0764 ± 0.0061	0.1398 ± 0.0038	0.0543 ± 0.0132	0.1396 ± 0.0043
	ARI	0.0453 ± 0.0007	0.0096 ± 0.0022	0.0936 ± 0.0182	0.0634 ± 0.0311	0.0859 ± 0.0106	0.0070 ± 0.0089	0.0222 ± 0.0040	0.0254 ± 0.0061	0.0207 ± 0.0068	0.0549 ± 0.0028
MotorImagery	ACC	0.5016 ± 0.0013	0.5360 ± 0.0147	0.5840 ± 0.0258	0.5048 ± 0.0072	0.5058 ± 0.0035	0.5111 ± 0.0059	0.5159 ± 0.0111	0.5127 ± 0.0048	0.5333 ± 0.0173	0.5508 ± 0.0070
	F1	0.3512 ± 0.0022	0.5084 ± 0.0319	0.5308 ± 0.0377	0.6586 ± 0.0706	0.3794 ± 0.0205	0.4565 ± 0.0416	0.5093 ± 0.0107	0.4401 ± 0.0338	0.5180 ± 0.0271	0.5362 ± 0.0175
	NMI	0.0003 ± 0.0002	0.0055 ± 0.0031	0.0540 ± 0.0291	0.0023 ± 0.0040	0.0011 ± 0.0008	0.0017 ± 0.0016	0.0012 ± 0.0011	0.0012 ± 0.0004	0.0044 ± 0.0042	0.0096 ± 0.0034
	ARI	-0.0002 ± 0.0000	0.0040 ± 0.0035	0.0254 ± 0.0160	-0.0003 ± 0.0004	-0.0003 ± 0.0001	-0.0010 ± 0.0011	-0.0010 ± 0.0014	-0.0006 ± 0.0001	0.0033 ± 0.0059	0.0082 ± 0.0028
NATOPS	ACC	0.6537 ± 0.0975	0.4861 ± 0.0512	0.3978 ± 0.0514	0.5672 ± 0.0859	0.4350 ± 0.0268	0.3267 ± 0.0919	0.3817 ± 0.0194	0.3522 ± 0.0811	0.2972 ± 0.0249	0.5022 ± 0.0157
	F1	0.6529 ± 0.0968	0.4643 ± 0.0539	0.4342 ± 0.1010	0.5167 ± 0.0816	0.4372 ± 0.0599	0.2057 ± 0.0868	0.3871 ± 0.0168	0.2131 ± 0.0691	0.2951 ± 0.0249	0.4958 ± 0.0103
	NMI	0.6454 ± 0.0805	0.4355 ± 0.0551	0.2990 ± 0.1190	0.5575 ± 0.0724	0.3896 ± 0.0459	0.2876 ± 0.1673	0.1897 ± 0.0207	0.4115 ± 0.1613	0.1097 ± 0.0324	0.5159 ± 0.0230
	ARI	0.5091 ± 0.1162	0.3056 ± 0.0590	0.1759 ± 0.0826	0.4179 ± 0.0983	0.2563 ± 0.0459	0.1603 ± 0.1191	0.1147 ± 0.0214	0.2551 ± 0.1261	0.0549 ± 0.0222	0.3489 ± 0.0284
PEMS-SF	ACC	0.4864 ± 0.0558	0.1968 ± 0.0065	0.4301 ± 0.0842	0.2268 ± 0.0179	0.3582 ± 0.0366	0.2514 ± 0.0897	0.4268 ± 0.0172	0.2377 ± 0.0102	0.4409 ± 0.0466	0.4436 ± 0.0844
	F1	0.5110 ± 0.0447	0.1941 ± 0.0064	0.4202 ± 0.0807	0.2263 ± 0.0204	0.3782 ± 0.0333	0.1185 ± 0.0734	0.4239 ± 0.0214	0.2370 ± 0.0154	0.4683 ± 0.0591	0.4902 ± 0.1198
	NMI	0.4510 ± 0.0416	0.0348 ± 0.0119	0.3687 ± 0.1015	0.1270 ± 0.0168	0.3336 ± 0.0281	0.2715 ± 0.2226	0.4593 ± 0.0196	0.1742 ± 0.0226	0.4167 ± 0.0441	0.4570 ± 0.1104
	ARI	0.2911 ± 0.0478	0.0030 ± 0.0034	0.2134 ± 0.0924	0.0374 ± 0.0156	0.1673 ± 0.0312	0.1333 ± 0.1105	0.2557 ± 0.0121	0.0651 ± 0.0082	0.2471 ± 0.0435	0.2839 ± 0.0947
PenDigits	ACC	0.4246 ± 0.0409	0.4128 ± 0.0672	0.4089 ± 0.0499	0.6548 ± 0.0159	0.3484 ± 0.0354	0.2670 ± 0.0505	0.4078 ± 0.0162	0.3647 ± 0.0150	0.4687 ± 0.0517	0.6841 ± 0.0284
	F1	0.4116 ± 0.0303	0.3971 ± 0.0746	0.4086 ± 0.0578	0.5693 ± 0.0171	0.3548 ± 0.0321	0.1710 ± 0.0688	0.3825 ± 0.0213	0.3747 ± 0.0150	0.4542 ± 0.0512	0.6938 ± 0.0411
	NMI	0.3233 ± 0.0199	0.3405 ± 0.0616	0.3994 ± 0.0617	0.6583 ± 0.0086	0.3059 ± 0.0165	0.2076 ± 0.0559	0.3390 ± 0.0131	0.2611 ± 0.0165	0.4348 ± 0.0630	0.6485 ± 0.0189
	ARI	0.2218 ± 0.0289	0.2151 ± 0.0560	0.2313 ± 0.0515	0.5196 ± 0.0193	0.1503 ± 0.0122	0.1033 ± 0.0425	0.2183 ± 0.0158	0.1528 ± 0.0112	0.2822 ± 0.0658	0.5305 ± 0.0276
PhonemeSpectra	ACC	0.1149 ± 0.0003	0.0633 ± 0.0006	0.0963 ± 0.0032	0.1124 ± 0.0038	0.1283 ± 0.0024	0.0454 ± 0.0059	0.0916 ± 0.0028	0.0871 ± 0.0023	0.0812 ± 0.0071	0.1127 ± 0.0015
	F1	0.1159 ± 0.0004	0.0605 ± 0.0010	0.0944 ± 0.0004	0.1048 ± 0.0067	0.1242 ± 0.0038	0.0099 ± 0.0028	0.0913 ± 0.0033	0.0845 ± 0.0017	0.0786 ± 0.0080	0.1126 ± 0.0017
	NMI	0.1793 ± 0.0023	0.0451 ± 0.0015	0.1386 ± 0.0089	0.2039 ± 0.0025	0.2085 ± 0.0038	0.0366 ± 0.0136	0.1090 ± 0.0036	0.1174 ± 0.0070	0.0932 ± 0.0191	0.1758 ± 0.0026
	ARI	0.0339 ± 0.0007	0.0038 ± 0.0003	0.0166 ± 0.0040	0.0424 ± 0.0048	0.0449 ± 0.0011	0.0046 ± 0.0026	0.0178 ± 0.0015	0.0186 ± 0.0020	0.0143 ± 0.0045	0.0334 ± 0.0008
RocketSports	ACC	0.4172 ± 0.0058	0.3921 ± 0.0150	0.4355 ± 0.0286	0.6205 ± 0.0400	0.4257 ± 0.0251	0.3129 ± 0.0296	0.3551 ± 0.0324	0.5036 ± 0.0475	0.3743 ± 0.0375	0.6007 ± 0.0721
	F1	0.4138 ± 0.0065	0.3898 ± 0.0135	0.4064 ± 0.0374	0.5330 ± 0.0327	0.4067 ± 0.0328	0.1215 ± 0.0508	0.3529 ± 0.0340	0.5273 ± 0.0345	0.3667 ± 0.0343	0.6006 ± 0.0707
	NMI	0.1398 ± 0.0042	0.1024 ± 0.0386	0.1138 ± 0.0264	0.4687 ± 0.0339	0.1250 ± 0.0289	0.0688 ± 0.0371	0.0601 ± 0.0348	0.3279 ± 0.0420	0.0818 ± 0.0394	0.3488 ± 0.0845
	ARI	0.0880 ± 0.0042	0.0590 ± 0.0167	0.0776 ± 0.0307	0.3723 ± 0.0445	0.0924 ± 0.0200	0.0127 ± 0.0191	0.0400 ± 0.0293	0.2394 ± 0.0441	0.0488 ± 0.0282	0.2585 ± 0.0965
SelfRegulationSCP1	ACC	0.5362 ± 0.0031	0.5447 ± 0.0307	0.7126 ± 0.0431	0.5027 ± 0.0000	0.6096 ± 0.0472	0.5537 ± 0.0342	0.5586 ± 0.0087	0.7340 ± 0.0059	0.5811 ± 0.0402	0.7754 ± 0.0128
	F1	0.5081 ± 0.0040	0.5434 ± 0.0324	0.7073 ± 0.0419	0.7065 ± 0.0000	0.6082 ± 0.0487	0.4723 ± 0.0964	0.5584 ± 0.0087	0.7184 ± 0.0076	0.5676 ± 0.0372	0.7745 ± 0.0125
	NMI	0.0059 ± 0.0009	0.0086 ± 0.0093	0.1642 ± 0.0748	0.0000 ± 0.0000	0.0421 ± 0.0290	0.0261 ± 0.0139	0.0102 ± 0.0031	0.2493 ± 0.0044	0.0328 ± 0.0305	0.2382 ± 0.0246
	ARI	0.0039 ± 0.0009	0.0100 ± 0.0128	0.1857 ± 0.0753	0.0000 ± 0.0000	0.0553 ± 0.0393	0.0154 ± 0.0124	0.0123 ± 0.0043	0.2182 ± 0.0111	0.0313 ± 0.0296	0.3028 ± 0.0274
SelfRegulationSCP2	ACC	0.5163 ± 0.0051	0.5258 ± 0.0162	0.5811 ± 0.0143	0.5027 ± 0.0000	0.5521 ± 0.0096	0.5347 ± 0.0210	0.5053 ± 0.0069	0.5432 ± 0.0054	0.5811 ± 0.0402	0.5253 ± 0.0141
	F1	0.4913 ± 0.0063	0.5242 ± 0.0166	0.5657 ± 0.0301	0.7065 ± 0.0000	0.5448 ± 0.0077	0.5302 ± 0.0239	0.5039 ± 0.0073	0.5113 ± 0.0110	0.5676 ± 0.0372	0.5163 ± 0.0084
	NMI	0.0011 ± 0.0006	0.0027 ± 0.0027	0.0234 ± 0.0043	0.0000 ± 0.0000	0.0090 ± 0.0034	0.0049 ± 0.0048	0.0002 ± 0.0004	0.0083 ± 0.0020	0.0328 ± 0.0305	0.0034 ± 0.0044
	ARI	-0.0010 ± 0.0006	0.0111 ± 0.0036	0.0224 ± 0.0083	0.0000 ± 0.0000	0.0088 ± 0.0040	0.0041 ± 0.0065	-0.0023 ± 0.0005	0.0056 ± 0.0018	0.0313 ± 0.0296	0.0009 ± 0.0038
SpokenArabicDigits	ACC	0.4902 ± 0.0258	0.1000 ± 0.0000	0.2415 ± 0.0313	0.3869 ± 0.0543	0.4655 ± 0.1073	0.1889 ± 0.0452	0.3692 ± 0.0053	0.1000 ± 0.0000	0.3054 ± 0.0040	0.4420 ± 0.0257
	F1	0.4939 ± 0.0288	0.0182 ± 0.0000	0.2320 ± 0.0343	0.3303 ± 0.0362	0.4617 ± 0.1044	0.1196 ± 0.0519	0.3713 ± 0.0084	0.1819 ± 0.0000	0.3030 ± 0.0060	0.4460 ± 0.0206
	NMI	0.4740 ± 0.0129	0.0000 ± 0.0000	0.1566 ± 0.0474	0.4331 ± 0.0779						

TABLE III
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART I. RED “+”
MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
ArticulatoryWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
			0.7155 ± 0.0137	0.7451 ± 0.0215	0.7824 ± 0.0132	0.5728 ± 0.0248
	✓		0.7739 ± 0.0437	0.7997 ± 0.0481	0.8255 ± 0.0189	0.6570 ± 0.0427
			0.6918 ± 0.0306	0.7336 ± 0.0245	0.7709 ± 0.0187	0.5517 ± 0.0369
	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
AtrialFibrillation	✓	✓	0.4867 ± 0.0163	0.4331 ± 0.0116	0.1050 ± 0.0047	0.0404 ± 0.0051
			0.4800 ± 0.0400	0.4681 ± 0.0854	0.1365 ± 0.0448	0.0664 ± 0.0421
	✓		0.4600 ± 0.0327	0.5024 ± 0.0485	0.1000 ± 0.0087	0.0346 ± 0.0071
	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
			0.9825 ± 0.0127	0.9824 ± 0.0129	0.9497 ± 0.0366	0.9533 ± 0.0338
BasicMotions	✓	✓	0.8900 ± 0.1317	0.9203 ± 0.0746	0.8924 ± 0.0754	0.8361 ± 0.1360
			0.9700 ± 0.0100	0.9698 ± 0.0102	0.9138 ± 0.0287	0.9202 ± 0.0263
	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
			0.7824 ± 0.0547	0.8177 ± 0.0436	0.8416 ± 0.0367	0.7078 ± 0.0644
	✓	✓	0.7371 ± 0.0227	0.7610 ± 0.0346	0.8038 ± 0.0198	0.6417 ± 0.0303
Character Trajectories			0.7369 ± 0.0590	0.7699 ± 0.0708	0.8138 ± 0.0468	0.6556 ± 0.0750
	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
			0.8489 ± 0.0349	0.8739 ± 0.0420	0.8737 ± 0.0444	0.7589 ± 0.0659
	✓	✓	0.7811 ± 0.0658	0.8191 ± 0.0588	0.8346 ± 0.0364	0.6817 ± 0.0702
			0.8078 ± 0.0114	0.8292 ± 0.0241	0.8376 ± 0.0130	0.6937 ± 0.0195
Cricket	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
			0.3700 ± 0.0179	0.3600 ± 0.0253	0.1143 ± 0.0234	0.0526 ± 0.0108
	✓	✓	0.3280 ± 0.0421	0.2966 ± 0.0425	0.0869 ± 0.0246	0.0196 ± 0.0205
			0.3680 ± 0.0194	0.3449 ± 0.0235	0.1229 ± 0.0145	0.0306 ± 0.0112
	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
ERing			0.6867 ± 0.0669	0.7380 ± 0.0341	0.6238 ± 0.0426	0.5008 ± 0.0669
	✓	✓	0.7300 ± 0.0485	0.7313 ± 0.0475	0.6198 ± 0.0400	0.5226 ± 0.0569
			0.7173 ± 0.0259	0.7082 ± 0.0253	0.6112 ± 0.0390	0.5033 ± 0.0415
	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
			0.4131 ± 0.0101	0.3546 ± 0.0347	0.2196 ± 0.0211	0.1907 ± 0.0156
EigenWorms	✓	✓	0.4425 ± 0.0451	0.4077 ± 0.0359	0.1829 ± 0.0256	0.1469 ± 0.0344
			0.3822 ± 0.0462	0.3397 ± 0.0392	0.1713 ± 0.0600	0.1368 ± 0.0716
	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
			0.4640 ± 0.0291	0.4648 ± 0.0300	0.2217 ± 0.0332	0.1511 ± 0.0259
	✓	✓	0.4720 ± 0.0368	0.4705 ± 0.0382	0.2330 ± 0.0451	0.1551 ± 0.0279
Epilepsy			0.4829 ± 0.0238	0.4847 ± 0.0243	0.2418 ± 0.0298	0.1637 ± 0.0212
	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
			0.3015 ± 0.0099	0.2777 ± 0.0082	0.0114 ± 0.0033	0.0049 ± 0.0030
	✓	✓	0.3088 ± 0.0183	0.2881 ± 0.0252	0.0184 ± 0.0066	0.0104 ± 0.0062
			0.2950 ± 0.0092	0.2797 ± 0.0108	0.0124 ± 0.0061	0.0045 ± 0.0039
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
			0.5051 ± 0.0028	0.5050 ± 0.0028	0.0001 ± 0.0001	0.0000 ± 0.0001
	✓	✓	0.5101 ± 0.0023	0.5092 ± 0.0027	0.0003 ± 0.0001	0.0003 ± 0.0002
			0.5063 ± 0.0076	0.5061 ± 0.0077	0.0003 ± 0.0005	0.0003 ± 0.0007
	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
FingerMovements			0.5168 ± 0.0101	0.5123 ± 0.0056	0.0013 ± 0.0015	0.0008 ± 0.0017
	✓	✓	0.5168 ± 0.0128	0.5125 ± 0.0160	0.0013 ± 0.0014	0.0005 ± 0.0019
			0.5130 ± 0.0074	0.5127 ± 0.0073	0.0006 ± 0.0007	0.0015 ± 0.0010
	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
			0.3051 ± 0.0154	0.3027 ± 0.0162	0.0139 ± 0.0036	0.0008 ± 0.0029
HandMovement Direction	✓	✓	0.3179 ± 0.0249	0.3148 ± 0.0246	0.0201 ± 0.0119	0.0049 ± 0.0111
			0.3060 ± 0.0126	0.3034 ± 0.0116	0.0144 ± 0.0029	0.0011 ± 0.0025
	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
			0.2102 ± 0.0132	0.2103 ± 0.0137	0.3030 ± 0.0188	0.0770 ± 0.0111
	✓	✓	0.2444 ± 0.0113	0.2460 ± 0.0156	0.3503 ± 0.0160	0.1050 ± 0.0109
Handwriting			0.2224 ± 0.0130	0.2234 ± 0.0113	0.3212 ± 0.0209	0.0852 ± 0.0100
	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
			0.6826 ± 0.0369	0.6002 ± 0.0271	0.0438 ± 0.0108	0.0952 ± 0.0277
	✓	✓	0.6841 ± 0.0410	0.5474 ± 0.0747	0.0328 ± 0.0106	0.0620 ± 0.0591
			0.7066 ± 0.0091	0.6463 ± 0.0947	0.0365 ± 0.0108	0.0846 ± 0.0516

reconstruction alone. The model begins to adjust view diversity and prototype alignment, so a temporary increase is expected before the route-level terms settle. In later epochs, DuckDuckGeese, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF show smoother refinement. EthanolConcentration has a wider variance band near the transition, but the curve also becomes controlled afterward. The additional curves support the staged training order by showing that route-level refinement does not cause persistent divergence.

D. Mask-Rate Scalability Analysis

The main paper reports runtime and GPU memory when N , T , and D increase. This subsection checks the source of the runtime saving more directly. If admission reduces encoder-side computation, then increasing the mask rate should lower normalized execution time because fewer patches enter attention. Fig. 3 tests this relation by varying N , T , and D separately.

TABLE IV
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART II. RED “+”
MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1351 ± 0.0015	0.1206 ± 0.0036	0.0147 ± 0.0008	0.0064 ± 0.0001
	✓	✓	0.1304 ± 0.0032	0.1115 ± 0.0076	0.0132 ± 0.0011	0.0048 ± 0.0013
			0.1323 ± 0.0039	0.1099 ± 0.0069	0.0139 ± 0.0017	0.0050 ± 0.0011
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8162 ± 0.0393	0.8557 ± 0.0416	0.8044 ± 0.0126	0.7321 ± 0.0429
	✓	✓	0.8750 ± 0.0347	0.8840 ± 0.0294	0.8220 ± 0.0239	0.7847 ± 0.0385
			0.8800 ± 0.0306	0.8802 ± 0.0270	0.8140 ± 0.0272	0.7648 ± 0.0553
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.2018 ± 0.0106	0.1853 ± 0.0156	0.1475 ± 0.0125	0.0561 ± 0.0074
	✓	✓	0.2096 ± 0.0131	0.1792 ± 0.0080	0.1458 ± 0.0102	0.0567 ± 0.0093
			0.2048 ± 0.0134	0.1824 ± 0.0031	0.1553 ± 0.0062	0.0589 ± 0.0052
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.2839 ± 0.0032	0.2979 ± 0.0079	0.3439 ± 0.0107	0.1012 ± 0.0057
	✓	✓	0.3272 ± 0.0278	0.3460 ± 0.0497	0.4212 ± 0.0300	0.1600 ± 0.0271
			0.2883 ± 0.0069	0.2959 ± 0.0148	0.3481 ± 0.0125	0.1084 ± 0.0080
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5360 ± 0.0178	0.5196 ± 0.0260	0.0053 ± 0.0044	0.0041 ± 0.0050
	✓	✓	0.5307 ± 0.0224	0.5249 ± 0.0284	0.0042 ± 0.0048	0.0032 ± 0.0066
			0.5238 ± 0.0115	0.4965 ± 0.0034	0.0035 ± 0.0038	0.0007 ± 0.0027
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4533 ± 0.0457	0.4550 ± 0.0494	0.4481 ± 0.0457	0.2797 ± 0.0460
	✓	✓	0.4528 ± 0.0511	0.4554 ± 0.0583	0.4444 ± 0.0859	0.2844 ± 0.0753
			0.4750 ± 0.0072	0.4797 ± 0.0059	0.4753 ± 0.0118	0.3082 ± 0.0133
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2768 ± 0.0192	0.2752 ± 0.0175	0.2469 ± 0.0372	0.1170 ± 0.0271
	✓	✓	0.4159 ± 0.0708	0.4722 ± 0.0819	0.4329 ± 0.0928	0.2465 ± 0.0908
			0.2550 ± 0.0164	0.2553 ± 0.0131	0.2111 ± 0.0293	0.0900 ± 0.0244
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.5957 ± 0.0208	0.5949 ± 0.0207	0.5736 ± 0.0164	0.4065 ± 0.0264
	✓	✓	0.6614 ± 0.0378	0.6457 ± 0.0436	0.6188 ± 0.0319	0.4953 ± 0.0431
			0.5954 ± 0.0262	0.5955 ± 0.0250	0.5702 ± 0.0197	0.4034 ± 0.0284
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		OOM	0.1036 ± 0.0047	0.1044 ± 0.0044	0.1576 ± 0.0107	0.0277 ± 0.0029
	✓	✓	0.1036 ± 0.0016	0.1027 ± 0.0021	0.1568 ± 0.0039	0.0289 ± 0.0014
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6330 ± 0.0454	0.6263 ± 0.0497	0.4037 ± 0.0372	0.3443 ± 0.0751
	✓	✓	0.5340 ± 0.0979	0.5100 ± 0.0966	0.3100 ± 0.1019	0.2449 ± 0.1091
			0.6026 ± 0.0949	0.6012 ± 0.0964	0.3912 ± 0.1121	0.3290 ± 0.1141
SelfRegulation SCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7647 ± 0.0254	0.7635 ± 0.0258	0.2224 ± 0.0410	0.2816 ± 0.0510
	✓	✓	0.7740 ± 0.0170	0.7730 ± 0.0160	0.2387 ± 0.0412	0.3002 ± 0.0376
			0.7647 ± 0.0103	0.7644 ± 0.0099	0.2150 ± 0.0213	0.2794 ± 0.0224
SelfRegulation SCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5279 ± 0.0107	0.5270 ± 0.0093	0.0026 ± 0.0022	0.0010 ± 0.0029
	✓	✓	0.5147 ± 0.0106	0.5141 ± 0.0105	0.0010 ± 0.0013	0.0013 ± 0.0017
			0.5142 ± 0.0032	0.5141 ± 0.0033	0.0006 ± 0.0003	0.0018 ± 0.0004
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.4712 ± 0.0196	0.4778 ± 0.0219	0.4407 ± 0.0153	0.2844 ± 0.0124
	✓	✓	0.4477 ± 0.0191	0.4502 ± 0.0231	0.4457 ± 0.0383	0.2802 ± 0.0395
			0.4637 ± 0.0203	0.4691 ± 0.0204	0.4426 ± 0.0119	0.2821 ± 0.0086
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.4000 ± 0.0277	0.3749 ± 0.0372	0.0458 ± 0.0276	0.0173 ± 0.0277
	✓	✓	0.4222 ± 0.0502	0.3944 ± 0.0506	0.0417 ± 0.0306	0.0215 ± 0.0349
			0.4148 ± 0.0544	0.3852 ± 0.0626	0.0547 ± 0.0419	0.0341 ± 0.0161
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.3373 ± 0.0086	0.3156 ± 0.0090	0.1918 ± 0.0200	0.0954 ± 0.0114
	✓	✓	0.3945 ± 0.0411	0.3881 ± 0.0581	0.2768 ± 0.0378	0.1487 ± 0.0261
			0.3127 ± 0.0232	0.2967 ± 0.0177	0.1782 ± 0.0133	0.0860 ± 0.0156

TABLE V
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART I. RED “+” MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics				
	Recon	Contra	ACC	F1	NMI	ARI	
ArticulatoryWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470	
	✓	✓	0.6035 ± 0.0317	0.6212 ± 0.0261	0.7016 ± 0.0278	0.4614 ± 0.0399	
	✓	✓	0.8247 ± 0.0562	0.8514 ± 0.0558	0.8747 ± 0.0426	0.7445 ± 0.0793	
	✓	✓	0.1419 ± 0.0485	0.2366 ± 0.0323	0.2485 ± 0.0639	0.0353 ± 0.0299	
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492	
	✓	✓	0.4933 ± 0.0249	0.4763 ± 0.0291	0.0833 ± 0.0242	0.0169 ± 0.0203	
	✓	✓	0.4733 ± 0.0389	0.4520 ± 0.0542	0.0765 ± 0.0312	0.0190 ± 0.0333	
	✓	✓	0.4600 ± 0.0389	0.4518 ± 0.0598	0.0937 ± 0.0356	0.0212 ± 0.0325	
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363	
	✓	✓	0.9050 ± 0.1226	0.9319 ± 0.0784	0.9161 ± 0.0911	0.8633 ± 0.1575	
	✓	✓	0.6600 ± 0.0366	0.8105 ± 0.0355	0.7754 ± 0.0499	0.6063 ± 0.0482	
	✓	✓	0.5100 ± 0.0289	0.6355 ± 0.0519	0.5518 ± 0.0478	0.3927 ± 0.0402	
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576	
	✓	✓	0.6714 ± 0.0832	0.7199 ± 0.0942	0.7443 ± 0.0645	0.5830 ± 0.0923	
	✓	✓	0.7389 ± 0.0708	0.7674 ± 0.0674	0.8189 ± 0.0503	0.6750 ± 0.0821	
	✓	✓	0.0651 ± 0.0000	0.1222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299	
	✓	✓	0.7900 ± 0.1095	0.8001 ± 0.1052	0.8279 ± 0.0699	0.6912 ± 0.1158	
	✓	✓	0.7511 ± 0.0708	0.7828 ± 0.0339	0.7682 ± 0.0476	0.5964 ± 0.0836	
	✓	✓	0.1000 ± 0.0333	0.2018 ± 0.0960	0.0721 ± 0.1442	0.0121 ± 0.0241	
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345	
	✓	✓	0.4740 ± 0.0546	0.4690 ± 0.0553	0.2439 ± 0.0457	0.1618 ± 0.0523	
	✓	✓	0.3880 ± 0.0306	0.3682 ± 0.0377	0.1562 ± 0.0375	0.0621 ± 0.0248	
	✓	✓	0.2000 ± 0.0000	0.3333 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760	
	✓	✓	0.7533 ± 0.0419	0.7553 ± 0.0421	0.6735 ± 0.0240	0.5854 ± 0.0353	
	✓	✓	0.8020 ± 0.0937	0.8006 ± 0.0951	0.7408 ± 0.0702	0.6688 ± 0.1070	
	✓	✓	0.1667 ± 0.0000	0.2857 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726	
	✓	✓	0.4556 ± 0.0523	0.4150 ± 0.0632	0.2774 ± 0.0369	0.2479 ± 0.0381	
	✓	✓	0.3382 ± 0.0317	0.3009 ± 0.0439	0.0843 ± 0.0295	0.0512 ± 0.0158	
	✓	✓	0.4247 ± 0.0000	0.5962 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336	
	✓	✓	0.5716 ± 0.0429	0.5746 ± 0.0356	0.3998 ± 0.0509	0.3182 ± 0.0499	
	✓	✓	0.4320 ± 0.0292	0.4259 ± 0.0276	0.2047 ± 0.0333	0.1442 ± 0.0279	
	✓	✓	0.4291 ± 0.0332	0.4217 ± 0.0355	0.1670 ± 0.0517	0.1224 ± 0.0370	
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123	
	✓	✓	0.2916 ± 0.0080	0.2841 ± 0.0078	0.0129 ± 0.0041	0.0032 ± 0.0014	
	✓	✓	0.3115 ± 0.0143	0.2799 ± 0.0252	0.0308 ± 0.0187	0.0144 ± 0.0106	
	✓	✓	0.3179 ± 0.0130	0.3277 ± 0.0507	0.0442 ± 0.0192	0.0238 ± 0.0127	
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004	
	✓	✓	0.5090 ± 0.0055	0.5086 ± 0.0056	0.0003 ± 0.0003	0.0003 ± 0.0003	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010	
	✓	✓	0.5135 ± 0.0097	0.5114 ± 0.0111	0.0008 ± 0.0009	0.0013 ± 0.0012	
	✓	✓	0.5231 ± 0.0102	0.5188 ± 0.0102	0.0019 ± 0.0012	0.0002 ± 0.0016	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052	
	✓	✓	0.3077 ± 0.0162	0.3061 ± 0.0171	0.0148 ± 0.0062	0.0005 ± 0.0053	
	✓	✓	0.3111 ± 0.0208	0.3077 ± 0.0221	0.0215 ± 0.0075	0.0053 ± 0.0072	
	✓	✓	0.3068 ± 0.0068	0.2901 ± 0.0134	0.0201 ± 0.0043	0.0021 ± 0.0020	
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180	
	✓	✓	0.1706 ± 0.0061	0.1741 ± 0.0083	0.2227 ± 0.0045	0.0385 ± 0.0019	
	✓	✓	0.2054 ± 0.0071	0.2071 ± 0.0059	0.2809 ± 0.0091	0.0663 ± 0.0054	
	✓	✓	0.1428 ± 0.0060	0.1408 ± 0.0057	0.1813 ± 0.0062	0.0217 ± 0.0024	
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404	
	✓	✓	0.5628 ± 0.0376	0.5053 ± 0.0802	0.0459 ± 0.0155	0.0065 ± 0.0412	
	✓	✓	0.7002 ± 0.0176	0.7537 ± 0.1522	0.0170 ± 0.0068	0.0161 ± 0.0034	
	✓	✓	0.7213 ± 0.0000	0.8381 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	

TABLE VI
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART II. RED “+” MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
	✓	✓	0.1206 ± 0.0063	0.1104 ± 0.0111	0.0053 ± 0.0027	0.0025 ± 0.0014
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
	✓	✓	0.8859 ± 0.0310	0.8942 ± 0.0270	0.8224 ± 0.0315	0.7627 ± 0.0522
	✓	✓	0.6441 ± 0.1450	0.6632 ± 0.1467	0.5833 ± 0.1473	0.4328 ± 0.1564
	✓	✓	0.1844 ± 0.0000	0.3113 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
	✓	✓	0.3362 ± 0.0017	0.2228 ± 0.0094	0.1699 ± 0.0142	0.0714 ± 0.0062
	✓	✓	0.1969 ± 0.0139	0.1776 ± 0.0102	0.1427 ± 0.0084	0.0537 ± 0.0075
	✓	✓	0.3155 ± 0.0000	0.4797 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
	✓	✓	0.3600 ± 0.0283	0.3572 ± 0.0231	0.4518 ± 0.0285	0.1916 ± 0.0259
	✓	✓	0.4161 ± 0.0210	0.4303 ± 0.0315	0.5402 ± 0.0261	0.2622 ± 0.0231
	✓	✓	0.2644 ± 0.0235	0.2673 ± 0.0222	0.3222 ± 0.0456	0.0927 ± 0.0307
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
	✓	✓	0.5079 ± 0.0044	0.4726 ± 0.0036	0.0004 ± 0.0004	0.0016 ± 0.0004
	✓	✓	0.5270 ± 0.0074	0.5048 ± 0.0179	0.0034 ± 0.0023	0.0009 ± 0.0017
	✓	✓	OOM	OOM	OOM	OOM
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
	✓	✓	0.4972 ± 0.0478	0.4940 ± 0.0434	0.4796 ± 0.0594	0.3245 ± 0.0508
	✓	✓	0.4328 ± 0.0428	0.4242 ± 0.0431	0.3578 ± 0.0955	0.2377 ± 0.0650
	✓	✓	0.1967 ± 0.0600	0.3526 ± 0.1337	0.0770 ± 0.1539	0.0221 ± 0.0441
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
	✓	✓	0.2368 ± 0.0216	0.2476 ± 0.0355	0.1705 ± 0.0511	0.0609 ± 0.0218
	✓	✓	0.2700 ± 0.0273	0.2785 ± 0.0369	0.1930 ± 0.0781	0.0771 ± 0.0412
	✓	✓	0.1477 ± 0.0000	0.2574 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
	✓	✓	0.7139 ± 0.0446	0.7030 ± 0.0448	0.6472 ± 0.0277	0.5420 ± 0.0420
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
	✓	✓	0.0985 ± 0.0017	0.0967 ± 0.0012	0.1443 ± 0.0051	0.0243 ± 0.0012
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
	✓	✓	0.6389 ± 0.0424	0.6436 ± 0.0345	0.4214 ± 0.0229	0.3526 ± 0.0483
	✓	✓	0.4488 ± 0.0172	0.4449 ± 0.0185	0.1929 ± 0.0490	0.1187 ± 0.0352
	✓	✓	0.4244 ± 0.0367	0.3947 ± 0.0725	0.1550 ± 0.0234	0.1061 ± 0.0245
Self RegulationSCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
	✓	✓	0.7458 ± 0.0132	0.7421 ± 0.0156	0.2011 ± 0.0274	0.2411 ± 0.0261
	✓	✓	0.7337 ± 0.0714	0.7246 ± 0.0859	0.1956 ± 0.0777	0.2376 ± 0.1041
	✓	✓	0.7480 ± 0.0175	0.7477 ± 0.0176	0.1878 ± 0.0261	0.2458 ± 0.0341
Self RegulationSCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
	✓	✓	0.5347 ± 0.0139	0.5325 ± 0.0152	0.0041 ± 0.0021	0.0030 ± 0.0029
	✓	✓	0.5158 ± 0.0173	0.5106 ± 0.0157	0.0017 ± 0.0031	0.0003 ± 0.0040
	✓	✓	0.5200 ± 0.0061	0.5191 ± 0.0059	0.0013 ± 0.0007	0.0009 ± 0.0009
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
	✓	✓	0.3631 ± 0.0512	0.3642 ± 0.0468	0.3212 ± 0.0486	0.1523 ± 0.0453
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
	✓	✓	0.4593 ± 0.0378	0.4537 ± 0.0361	0.0530 ± 0.0194	0.0194 ± 0.0251
	✓	✓	0.4519 ± 0.0363	0.4355 ± 0.0497	0.0543 ± 0.0236	0.0178 ± 0.0199
	✓	✓	0.4296 ± 0.0181	0.3740 ± 0.0214	0.0924 ± 0.0506	0.0016 ± 0.0243
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
	✓	✓	0.4023 ± 0.0342	0.3992 ± 0.0359	0.2887 ± 0.0371	0.1684 ± 0.0280
	✓	✓	0.6318 ± 0.0456	0.6520 ± 0.0381	0.5779 ± 0.0160	0.4497 ± 0.0166
	✓	✓	0.1250 ± 0.0000	0.2222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000

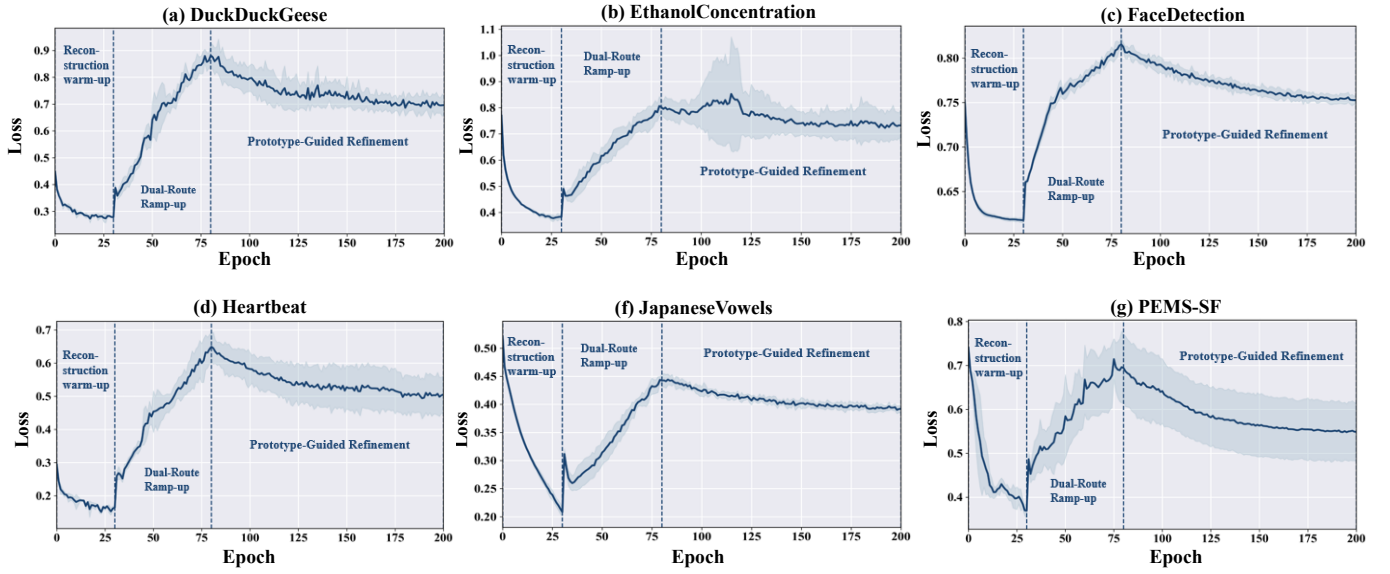


Fig. 2. Additional loss-evolution curves during SIGMA training on six UEA datasets. Each curve reports the mean loss across five random seeds, and the shaded band denotes standard deviation. Panels (a) to (f) correspond to DuckDuckGeese, EthanolConcentration, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF, respectively. Vertical dashed lines mark the reconstruction-only, dual-route, and prototype-guided refinement stages.

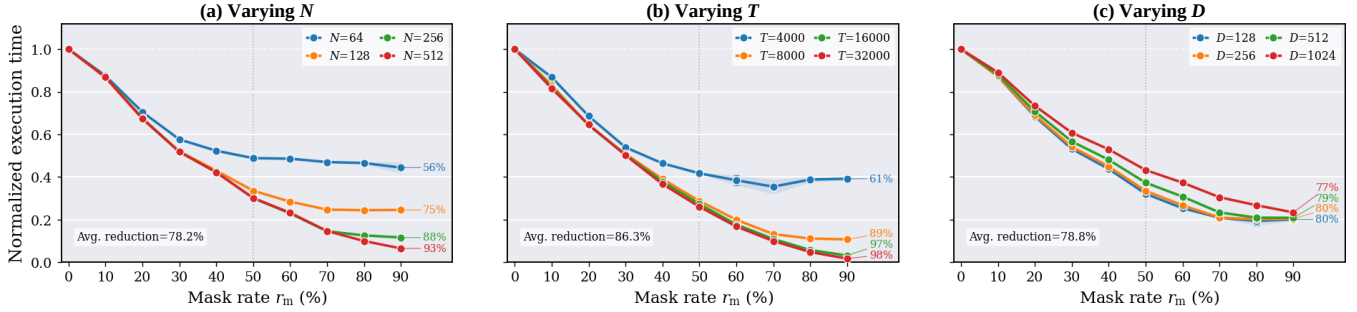


Fig. 3. Scalability under increasing mask rates. The three panels vary N , T , and D , respectively. For each setting, execution time is normalized by the cost at zero mask rate. Shaded bands denote standard deviation, and endpoint annotations report the reduction at 90% mask rate.

30 UEA datasets. Architecture and route-internal ablations expand Q2, larger-scale and mask-rate tests examine Q3, and additional robustness diagnostics complement Q4 under noise perturbation and embedding compression. The supplementary results therefore make the main claims traceable without introducing another evaluation setting.

REFERENCES

- [1] C. Wang, M. Du, X. Jiang, and Y. Dong, “Fuzzy cluster-aware contrastive clustering for time series,” *Pattern Recognition*, vol. 173, p. 112899, 2026.
- [2] C. Wang, M. Du, and X. Jiang, “Time-frequency augmented multi-level contrastive clustering for time series,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 31, 2026, pp. 26 142–26 150.
- [3] C. He, X. Huang, G. Jiang, Z. Li, D. Lian, H. Xie, E. Chen, X. Liang, Z. Zheng, and P. P. C. Lee, “GTM: A general time-series model for enhanced representation learning of time-series data,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. [Online]. Available: <https://openreview.net/forum?id=PWM6FERWz9>
- [4] Z. Tan, X. Luo, Y. Liu, and Y. Zhang, “Mask the redundancy: Evolving masking representation learning for multivariate time-series clustering,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 30, 2026, pp. 25 787–25 795.
- [5] E. Fu and Y. Hu, “Frequency-masked embedding inference: A non-contrastive approach for time series representation learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 39, no. 16, 2025, pp. 16 639–16 647.
- [6] Y. Li, M. Du, X. Jiang, and N. Zhang, “Contrastive learning-based multi-view clustering for incomplete multivariate time series,” *Information Fusion*, vol. 117, p. 102812, 2025.
- [7] P. Boniol, D. Tiano, A. Bonifati, and T. Palpanas, “ k -graph: A graph embedding for interpretable time series clustering,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 5, pp. 2680–2694, 2025.
- [8] J. Liu and S. Chen, “Timesurl: Self-supervised contrastive learning for universal time series representation learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 38, no. 12, 2024, pp. 13 918–13 926.
- [9] S. Gao, T. Koker, O. Queen, T. Hartvigsen, T. Tsiligkaridis, and M. Zitnik, “Units: A unified multi-task time series model,” in *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 140 589–140 631.

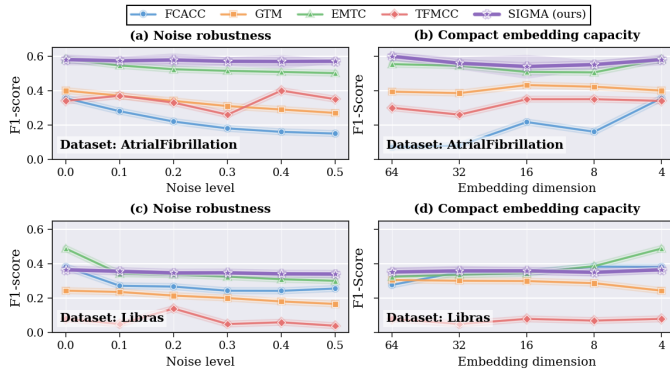


Fig. 4. **Robustness and compactness diagnostics.** F1-score on AtrialFibrillation and Libras under (a) increasing Gaussian noise and (b) reduced embedding dimensions from 64 to 4.